

Methods

TReSpire – a biophysical TRee Stem respiration model

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Summary

- Mechanistic models of plant respiration remain poorly developed, especially in stems and woody tissues where measurements of CO₂ efflux do not necessarily reflect local respiratory activity.
- We built a process-based model of stem respiration that couples water and carbon fluxes at the organ level (TReSpire). To this end, sap flow, stem diameter variations, xylem and soil water potential, stem temperature, stem CO₂ efflux and nonstructural carbohydrates were measured in a maple tree, while xylem CO₂ concentration and additional stem and xylem diameter variations were monitored in an ancillary tree for model validation.
- TReSpire realistically described: (1) turgor pressure to differentiate growing from nongrowing metabolism; (2) maintenance expenditures in xylem and outer tissues based on Arrhenius kinetics and nitrogen content; and (3) radial CO₂ diffusivity and CO₂ solubility and transport in the sap solution. Collinearity issues with phloem unloading rates and sugar–starch interconversion rates suggest parallel submodelling to close the stem carbon balance.
- TReSpire brings a breakthrough in the modelling of stem water and carbon fluxes at a detailed (hourly) temporal resolution. TReSpire is calibrated from a sink-driven perspective, and has potential to advance our understanding on stem growth dynamics, CO₂ fluxes and underlying respiratory physiology across different species and phenological stages.

Introduction

As trees grow, biomass is mostly allocated to stems and branches, which constitute the largest living sink of anthropogenic CO₂ emissions at the global scale (Cuny *et al.*, 2015). To fulfil stem metabolic activities, assimilated carbon (C) is partially consumed via respiratory processes, and CO₂ is emitted back to the atmosphere as a by-product. Therefore, despite uncertainty in estimates of stem respiration (R_S) at large spatial scales, R_S is expected to play a major role in ecosystem C balances, with stem CO₂ efflux to the atmosphere (E_A) accounting for 5–35% of ecosystem respiration (Capioli *et al.*, 2016; Salomón *et al.*, 2017), and being 11% and 20% of the amount of ecosystem gross and net primary production (GPP and NPP, respectively) (Yang *et al.*, 2016). Nevertheless, our understanding of stem and woody tissue respiration is deficient, especially when compared with our knowledge of CO₂ exchange at the leaf level (Meir *et al.*, 2017; Fatichi *et al.*, 2019). This is clearly illustrated by how woody tissue respiration is estimated by leading terrestrial biosphere models (TBMs). Most TBMs partition whole-plant (and hence woody tissue) respiration into two components according to the growth and maintenance respiration paradigm (GMRP) postulated in the early 1970s (reviewed by Atkin *et al.*, 2017). Maintenance respiration (R_M) is commonly extrapolated from estimates of leaf dark respiration to stem and roots according to N allocation patterns across tree

organs and assuming the same N– R_M relationship for leaves, stem and roots, despite evidence challenging this approach (Reich *et al.*, 2008). Growth respiration (R_G) is assumed to be a fixed fraction of NPP or GPP minus R_M . The latter assumption simplifies quantification of the respiratory costs associated with synthesis of new tissues, whose parametrisation is responsible for the largest source of uncertainty of NPP in TBMs (c. 50% in deciduous hardwood plant functional types and 30% in evergreen and non-woody ones; Dietze *et al.*, 2014). Overall, the fraction of autotrophic respiration that occurs in stems and roots (22% and 38%, respectively, according to an interbiome comparison; see Capioli *et al.*, 2016) is ultimately derived from our better understanding of leaf respiratory metabolism, despite respiratory differences between photosynthetic and nonphotosynthetic tissues (Armstrong *et al.*, 2006) and organs (Reich *et al.*, 2008).

Methodological feasibility has largely constrained our knowledge on stem (and woody tissue) respiration. While leaf photosynthesis and respiration can be directly estimated from measurements of CO₂ exchange in (apical) leaves enclosed in sealed chambers, CO₂ efflux from the stem does not necessarily reflect the respiratory activity of underlying tissues (Teskey *et al.*, 2008; Trumbore *et al.*, 2013). Respired CO₂ in the stem can radially diffuse to the atmosphere, axially diffuse along the stem in the gas phase (De Roo *et al.*, 2019), be fixed by anaplerotic reactions or woody tissue photosynthesis (Ávila *et al.*, 2014;

Hilman *et al.*, 2018), or dissolve in the sap solution and be transported upward with the transpiration stream, which likely confound the interpretation of E_A measurements to the greatest extent (reviewed by Teskey *et al.*, 2008, 2017). To account for diverse fates of respired CO_2 in the stem, several methodological approaches have been proposed (McGuire & Teskey, 2004; Angert *et al.*, 2012; Salomón *et al.*, 2019a). However, there is not a universal and well established protocol to bring these together and measure R_S in a way that could facilitate comparison among experiments. Not surprisingly, there is no consensus on the scalar (surface, volume or mass) in which R_S and E_A should be expressed (Meir *et al.*, 2017), nor on their terminology. In the context of climate change, some studies have shown downregulation of R_S in drought-stressed trees (see Rodríguez-Calcerrada *et al.*, 2014; and references therein), whereas contrasting results (see Rowland *et al.*, 2018; and references therein) evidence our deficient understanding of the effect of drought on woody tissue respiration. Likewise, elevated atmospheric CO_2 can result in a positive, neutral or negative effect on R_S (see Salomón *et al.*, 2019b; and references therein). So far, there are no theoretical frameworks that could reconcile disparity among these context-dependent observations and make consistent predictions of stem and woody tissue respiration under changing climate regimes.

From a biochemical perspective, CO_2 production encompasses a wide array of metabolic processes difficult to quantify (Sweetlove *et al.*, 2013; Tcherkez, 2017), and therefore 'plant respiratory regulation is too complex for a mechanistic representation in current terrestrial productivity models for carbon accounting and global change research' (Gifford, 2003). We are not yet able to model whole-tree respiration at the same biochemical detail as we do for photosynthesis via the well known Farquhar *et al.* (1980) model (Atkin *et al.*, 2014). Nevertheless, we do have the basis to build biophysical models of R_S at the organ level. Several models simulate stem hydraulic functioning (Steppe *et al.*, 2006, 2008; Chan *et al.*, 2016; Mencuccini *et al.*, 2017; Coussement *et al.*, 2018), phloem transport (Thompson & Holbrook, 2003; Minchin & Lacomte, 2017), and coupled water and C fluxes along the stem (Lacomte & Minchin, 2008; Hölttä *et al.*, 2009; De Schepper & Steppe, 2010). These approaches provide key information on the hydric and osmotic status of stem living cells that regulate their respiratory metabolism. Some of these models simulate cell turgor pressure, a critical driver of cell growth (Steppe *et al.*, 2006, 2008, 2015; Fatichi *et al.*, 2014; Coussement *et al.*, 2018), and further separate irreversible plastic stem growth from elastic fluctuations in bark water content (Steppe *et al.*, 2006; Mencuccini *et al.*, 2017). This is an essential step to disentangle growth from maintenance metabolism on a subdaily basis (Saveyn *et al.*, 2007; Steppe *et al.*, 2015; Salomón *et al.*, 2018), and to consider plant respiration within sink-source theoretical frameworks (Atkin *et al.*, 2014; Fatichi *et al.*, 2019). By combining these process-based models with knowledge on thermal regulation (Ryan, 1991; Tjoelker *et al.*, 2001) and nitrogen (N) dependency (Ryan, 1991; Reich *et al.*, 2008) of respiratory processes within the GMRP (Amthor, 2000), it is feasible to model R_S at

the organ level, and simulate diffusion and solubility of respired CO_2 within woody tissues according to CO_2 gradients along the xylem-atmosphere continuum and equilibrium reactions at the gas-liquid interface (McGuire & Teskey, 2002; Hölttä & Kolari, 2009). Furthermore, stem C sources (phloem unloading, starch depolymerisation, woody tissue photosynthesis and anaplerotic fixation) fuelling C sinks (respiration, growth, starch synthesis and osmoregulation) could be quantified to close the stem C balance (Lacomte & Minchin, 2008; De Schepper & Steppe, 2010) and eventually incorporate biophysical processes into whole-plant and large-scale models.

The main goal of this work is to develop a TReStem respiration model (TReSpire). To this end, R_S and associated respiratory fluxes were represented by process-based equations integrating: stem allometry; function and location of different woody tissues; stem water and C fluxes; cell turgor to differentiate growing from nongrowing metabolism on a subdaily basis; temperature driven Arrhenius kinetics; N-dependency of respiratory processes; CO_2 diffusivity in the gas phase and its solubility in the sap solution; and C sources fuelling respiratory metabolism and additional stem C sinks. We pursue building a mechanistic model able to realistically describe R_S under varying biotic and abiotic conditions to advance knowledge in the respiratory physiology of woody tissues.

Materials and Methods

Description

Model overview TReSpire describes geometrically the stem as two coaxial cylinders separated by the cambial layer: the xylem compartment and the outer tissues compartment, which integrates phloem and bark tissues (Génard *et al.*, 2001; Steppe *et al.*, 2006). Briefly, TReSpire is composed of three submodels (Fig. 1). The first submodel is a refined version of the 'water flow and storage' model, which describes stem hydraulic functioning (Steppe *et al.*, 2006, 2008). Here, the transpiration demand creates a water potential gradient ($\Delta\Psi$) along the xylem that results in a radial $\Delta\Psi$ between xylem and outer tissues. This gradient pulls water from the outer tissues to the xylem to fulfil transpiration needs resulting in stem shrinkage. The opposite flow occurs when transpiration decreases and stem water reservoirs are replenished resulting in stem swelling. This submodel differentiates the osmotic and turgor pressure components of Ψ in outer tissues. When turgor is higher than a threshold value for irreversible plastic growth, stem growth occurs. Otherwise, stem diameter variations are uniquely ascribed to elastic fluctuations due to changes in water content. Stem plastic growth is a driver of the second submodel 'CO₂ sources and fluxes'. The theoretical framework of the GMRP states that respiration is partitioned into two components: (1) growth respiration is directly related to stem growth rates; (2) maintenance respiration occurs in both xylem and outer tissues, is temperature dependent according to Arrhenius kinetics, and scales with N content of the living/respinding tissues (including parenchyma in xylem). The submodel further accounts for radial CO_2 diffusion through the xylem and outer tissues, xylem

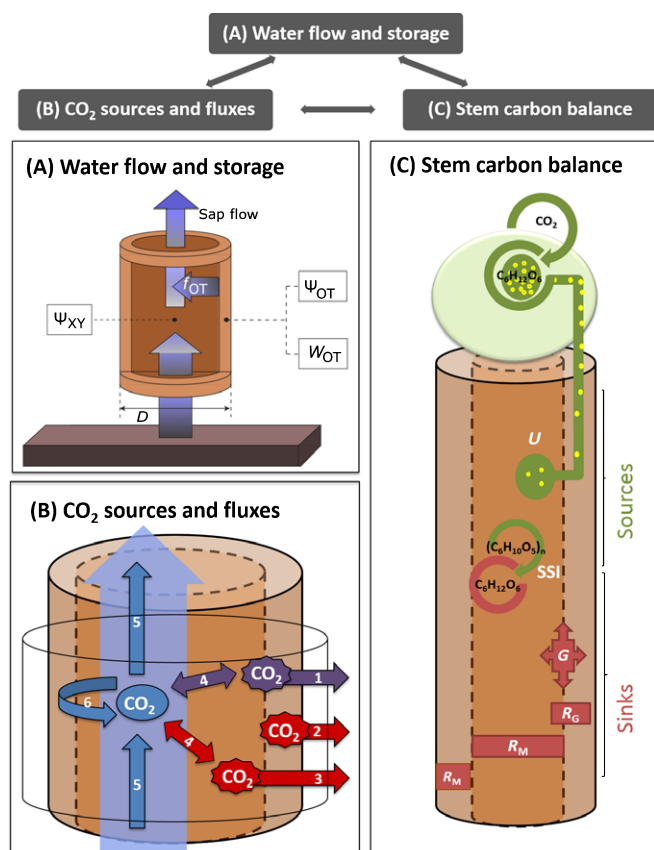


Fig. 1 Schematic of submodels integrating TRESpire. The 'water flow and storage' submodel (A) describes the hydraulic functioning of the stem. Driven by sap flow and soil water potential, and calibrated against diameter (D) variations and xylem water potential (Ψ_{XY}), it simulates stem irreversible plastic growth and water content and potential of outer tissues (W_{OT} and Ψ_{OT}). The 'CO₂ sources and fluxes' submodel (B) is driven by stem temperature and plastic growth, and is calibrated against stem CO₂ efflux. It simulates growth respiration in the cambium (in purple) and maintenance respiration of xylem and outer tissues (in red). Respirated CO₂ either radially effluxes to the atmosphere (1–3), dissolves in the sap solution (4) and is transported axially through the xylem (5), or is stored within the stem in gaseous and liquid phases (6). The 'stem carbon balance' submodel (C) is driven by previous outputs and calibrated against the vertical gradient in stem D variations and the concentration of nonstructural carbohydrates in the stem. It simulates stem carbon sources and sinks (in green and red, respectively): phloem unload (U), sugar–starch interconversion (SSI), stem growth (G), growth and maintenance respiration (R_G and R_M , respectively), and osmoregulatory processes.

CO₂ transport through the xylem, and CO₂ storage in both compartments. Finally, the third submodel 'Stem carbon balance' quantifies C sources and sinks to close the C balance at the organ level. See Tables 1 and 2 for the description, acronym and unit of model variables and parameters, respectively.

Water flow and storage submodel ('submodel A') It is based on the original model of stem hydraulic functioning (Steppe *et al.*, 2006, 2008) with two main refinements: (1) thickness of the outer tissues compartment is represented by a linear relationship derived from simultaneous measurements of xylem and stem diameter variations (ΔD) instead of a nonlinear allometric

relationship (cf. Génard *et al.*, 2001); and (2) elastic osmotically driven changes are added to elastic tension driven and inelastic growth changes to estimate diameter stem variations (Mencuccini *et al.*, 2013, 2017; Chan *et al.*, 2016). Note that the theoretical background and mathematical development of this submodel are explained elsewhere and uniquely model refinements are explained in detail here.

Root water potential (Ψ_{ROOT} , MPa) is estimated from soil water potential (Ψ_{SOIL} , MPa):

$$\Psi_{\text{ROOT}} = \Psi_{\text{SOIL}} + D_{\text{SOIL}}, \quad \text{Eqn 1}$$

where D_{SOIL} (MPa) accounts for the disequilibrium between Ψ_{SOIL} and predawn Ψ_{ROOT} . Water exchange between the xylem and outer tissues (f_{OT} , g h^{−1}) is the time derivative of the water content in outer tissues (W_{OT} , g), which is estimated from Ohm's law considering that sap flow through the xylem (SF , g h^{−1}) results from the combination of f_{OT} and root water uptake (F_{SOIL} , g h^{−1}) (cf. Eqn 9 in Steppe *et al.*, 2006):

$$\frac{dW_{\text{OT}}}{dt} = \frac{\Psi_{\text{ROOT}} - \Psi_{\text{OT}} - (SF R_{XY})}{R_{XY} + R_{\text{OT}}}, \quad \text{Eqn 2}$$

$$f_{\text{OT}} = \frac{dW_{\text{OT}}}{dt}, \quad \text{Eqn 3}$$

where Ψ_{OT} (MPa) is Ψ in outer tissues, R_{XY} (MPa h g^{−1}) is the xylem axial resistance to water flow, and R_{OT} (MPa h g^{−1}) is the radial resistance to water exchange between xylem and outer tissues. Stem hydraulic capacitance of the outer tissues (C_{OT} , g MPa^{−1}) is defined by the ratio between W_{OT} and Ψ_{OT} , and the time derivative of Ψ_{OT} is estimated as follows:

$$\frac{d\Psi_{\text{OT}}}{dt} = \frac{f_{\text{OT}}}{C_{\text{OT}}}. \quad \text{Eqn 4}$$

Xylem water potential (Ψ_{XY} , MPa) is given by:

$$\Psi_{XY} = f_{\text{OT}} R_{\text{OT}} + \Psi_{\text{OT}}, \quad \text{Eqn 5}$$

and F_{SOIL} is calculated from Ohm's law:

$$F_{\text{SOIL}} = \frac{\Psi_{\text{ROOT}} - \Psi_{XY}}{R_{XY}}. \quad \text{Eqn 6}$$

Changes in W_{OT} are linked to the turgor pressure of outer tissues (Ψ_{OT}^p , MPa):

$$\frac{d\Psi_{\text{OT}}^p}{dt} = \frac{dW_{\text{OT}}}{dt} \frac{\epsilon}{\rho_W V_{\text{OT}}}, \quad \text{Eqn 7}$$

where ρ_W (g m^{−3}) is water density, V_{OT} (m³) is the volume of the outer tissues compartment, and ϵ (MPa) is the bulk elastic modulus of outer tissues in relation to reversible dimensional changes, which is proportional to stem diameter (D , m) and Ψ_{OT}^p :

Table 1 Acronym, unit and description of model variables.

	Acronym	Unit	Description
Water flow and storage	Ψ_{SOIL}	MPa	Soil water potential
	Ψ_{ROOT}	MPa	Root water potential
	Ψ_{XY}	MPa	Water potential in xylem
	Ψ_{OT}	MPa	Water potential in outer tissues
	$\Psi_{\text{OT}}^{\text{P}}$	MPa	Pressure component of Ψ_{OT}
	$\Psi_{\text{OT}}^{\text{O}}$	MPa	Osmotic component of Ψ_{OT}
	SF	g h^{-1}	Sap flow through the xylem
	f_{OT}	g h^{-1}	Water exchange between xylem and outer tissues
	F_{SOIL}	g h^{-1}	Root water uptake
	W_{OT}	g	Water content in outer tissues
	V_{OT}	m^3	Volume of the outer tissues
	ε	MPa	Bulk elastic modulus related to elastic changes
	D	m	Stem diameter
	D_{XY}	m	Xylem diameter
	d_{OT}	m	Thickness of the outer tissues
	dD_{P}/dt	m h^{-1}	Plastic growth changes in D
	dD_{ET}/dt	m h^{-1}	Elastic tension driven changes in D
	dD_{EO}/dt	m h^{-1}	Elastic osmotically driven changes in D
CO ₂ sources and fluxes	G	$\text{mol C m}^{-3} \text{h}^{-1}$	Stem growth
	R_{G}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	Growth respiration
	R_{M_XY}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	Maintenance respiration in the xylem
	R_{M_OT}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	Maintenance respiration in outer tissues
	R_{S}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	Stem respiration
	T	$^{\circ}\text{C}$ or $^{\circ}\text{K}^1$	Stem temperature
	V_{XY}	m^3	Volume of xylem within the stem segment
	V_{S}	m^3	Volume of the stem segment
	A_{XY}	m^2	Axial surface of xylem within the stem segment
	A_{S}	m^2	Axial surface of the stem segment
	k_{H}	$\text{mol atm}^{-1} \text{L}^{-1}$	Henry's law coefficient for CO ₂ in water
	k_1		First acidity constant
	k_2		Second acidity constant
	$[\text{CO}_2]_{\text{XY}}$	$\text{mol CO}_2 \text{m}^{-3}$	CO ₂ concentration in xylem (gas phase)
	$[\text{CO}_2]_{\text{OT}}$	$\text{mol CO}_2 \text{m}^{-3}$	CO ₂ concentration in outer tissues (gas phase)
	$[\text{CO}_2^*]$	$\text{mol CO}_2 \text{L}^{-1}$	CO ₂ concentration dissolved in sap solution
	E_{XY}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ efflux from xylem to outer tissues
	F_{T}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ transport through xylem
	ΔS_{XY}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ storage flux in the xylem in gaseous phase
	ΔS^*_{XY}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ storage flux in the xylem in liquid phase
	E_{A}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ efflux from xylem to atmosphere

Table 1. (Continued)

	Acronym	Unit	Description
Stem C balance	ΔS_{OT}	$\text{mol CO}_2 \text{m}^{-3} \text{h}^{-1}$	CO ₂ storage flux in outer tissues in gaseous phase
	Dif	$\text{m}^2 \text{h}^{-1}$	Diffusivity
	VGF	$\text{m}^3 \text{m}^{-3}$	Volumetric gas fraction
	VWF	$\text{m}^3 \text{m}^{-3}$	Volumetric water fraction
	$\Delta[\text{CO}_2^*]$	$\text{mol CO}_2 \text{m}^{-3}$	Vertical gradient in $[\text{CO}_2^*]$
	$\Psi_{\text{OT UP}}^{\text{P}}$	MPa	$\Psi_{\text{OT}}^{\text{P}}$ in the upper part of the stem ²
	$\Delta \Psi_{\text{OT}}^{\text{P}}$	MPa	Pressure drop along the stem
	$\Delta[\text{SU}]_{\text{OT}}$	g m^{-3}	Drop in $[\text{SU}]$ in outer tissues along the stem
	U	$\text{g m}^{-3} \text{h}^{-1}$	Phloem unload
	$[\text{SU}]$	g m^{-3}	Sucrose concentration (in outer tissues)
	$[\text{ST}]$	g m^{-3}	Starch concentration (in xylem)
	SSI	$\text{g m}^{-3} \text{h}^{-1}$	Sugar–starch interconversion
	G_{SU}	$\text{g m}^{-3} \text{h}^{-1}$	Growth sink (in sucrose equivalents)
	R_{S_SU}	$\text{g m}^{-3} \text{h}^{-1}$	Stem respiration sink (in sucrose equivalents)

Some variables are used in more than one submodel. Classification in this table is made according to their first appearance in the model.

¹Temperature unit depends on the equation.

²Submodel 'water flow and storage' is partially run for a second time for the upper part of the stem to estimate $\Psi_{\text{OT UP}}^{\text{P}}$, in addition to ε_{UP} , $V_{\text{OT UP}}$, $\Psi_{\text{OT UP}}^{\text{P}}$, $D_{\text{XY UP}}$, $d_{\text{OT UP}}$ and D_{UP} .

$$\varepsilon = \varepsilon_0 D \Psi_{\text{OT}}^{\text{P}}, \quad \text{Eqn 8}$$

where ε_0 (m^{-1}) is the proportionality constant. Assuming the thickness of the outer tissues (d_{OT} , m) is much smaller than the xylem diameter (D_{XY} , m), volume of outer tissues (V_{OT}) is estimated as:

$$V_{\text{OT}} = \pi d_{\text{OT}} D_{\text{XY}} L, \quad \text{Eqn 9}$$

where L (m) is the length of the stem segment. The osmotic pressure in the outer tissues ($\Psi_{\text{OT}}^{\text{O}}$, MPa) is the difference between $\Psi_{\text{OT}}^{\text{P}}$ and Ψ_{OT} ; and their time derivatives are estimated likewise:

$$\frac{d\Psi_{\text{OT}}^{\text{O}}}{dt} = \frac{d\Psi_{\text{OT}}^{\text{P}}}{dt} - \frac{d\Psi_{\text{OT}}}{dt}. \quad \text{Eqn 10}$$

Temporal changes in stem diameter (dD/dt , m h^{-1}) are calculated as the additive effect of irreversible plastic growth (dD_{P}/dt , m h^{-1}), elastic tension-driven (dD_{ET}/dt , m h^{-1}) and elastic osmotically driven (dD_{EO}/dt , m h^{-1}) changes in water content of outer tissues (Chan *et al.*, 2016; Mencuccini *et al.*, 2017):

$$\frac{dD}{dt} = \frac{dD_{\text{P}}}{dt} + \frac{dD_{\text{ET}}}{dt} + \frac{dD_{\text{EO}}}{dt}. \quad \text{Eqn 11}$$

When $\Psi_{\text{OT}}^{\text{P}}$ exceeds a threshold value for cell growth (Γ , MPa), irreversible plastic cell growth occurs (cf. Eqns 28, 29 in Steppe *et al.*, 2006):

Table 2 Acronym, unit and description of model parameters and constants.

	Acronym	Unit	Description
Water flow and storage	D_{SOIL}	MPa	Disequilibrium between soil and root Ψ
	R_{XY}	MPa h g ⁻¹	Xylem resistance to axial water flow
	R_{OT}	MPa h g ⁻¹	Radial resistance to water flow
	C_{OT}	g MPa ⁻¹	Capacitance of OT
	ρ_{W}	g m ⁻³	Water density
	ε_0	m ⁻¹	Proportionality constant
	L	m	Stem length
	Γ	MPa	Threshold value for cell growth
	Φ	MPa ⁻¹ h ⁻¹	Extensibility of cell walls related to plastic growth
	k_d	m m ⁻¹	Change in OT thickness relative to stem D
CO ₂ sources and fluxes	a		Allometric parameter
	b		Allometric parameter
	ρ_{OT}	g m ⁻³	Density of outer tissues
	ρ_{XY}	g m ⁻³	Density of xylem
	C_{C}	g C g ⁻¹	Stem carbon content (on a dry mass basis)
	M_{C}	g mol ⁻¹	Carbon molar mass
	M_{N}	g mol ⁻¹	Nitrogen molar mass
	Y_{G}	mol C mol C ⁻¹	Growth yield
	$R_{\text{b-N}}$	mol CO ₂ mol N ⁻¹ h ⁻¹	Basal respiration at a reference $T(T_{\text{b}})$
	T_{b}	°C	Reference temperature
C balance	N_{XY}	g N g ⁻¹	Nitrogen concentration in xylem
	N_{OT}	g N g ⁻¹	Nitrogen concentration in outer tissues
	Q_{10}		Temperature sensitivity of maintenance R
	pH		Sap pH
	R	m ³ atm K ⁻¹ mol ⁻¹	Ideal gas constant (submodel B)
		m ³ MPa K ⁻¹ mol ⁻¹	Ideal gas constant (submodel C)
	$[\text{CO}_2]_{\text{ATM}}$	mol CO ₂ m ⁻³	Atmospheric CO ₂ concentration
	Dif_0	m ² h ⁻¹	Diffusivity at zero VGF
	Dif_e		Exponential parameter of Dif
	$(\text{VWF})_{t0}$	m ³ m ⁻³	Initial volumetric water fraction
	VC_{WF}	m ³ m ⁻³	Volumetric cell wall fraction
	$(W_{\text{OT}})_{t0}$	g	Initial water content in outer tissues
	L_{S}	m	Length of monitored stem segment
	$\Delta[\text{CO}_2^*]_{\text{slope}}$	mol CO ₂ m ⁻³ h ⁻¹	Time derivative of $\Delta[\text{CO}_2^*]$
	g	MPa m ² g ⁻¹	Gravitational constant
	M_{SU}	g mol ⁻¹	Sucrose molar mass
	V_{max}	mol m ⁻³ h ⁻¹	Michaelis–Menten kinetic parameter
	k_{M}	mol m ⁻³	Michaelis–Menten kinetic parameter

Table 2. (Continued)

Acronym	Unit	Description
k_{starch}	h ⁻¹	Kinetic parameter of sugar–starch interconversion
$[\text{SU}]_{\text{targ}}$	g m ⁻³	Target sucrose concentration

Some parameters and constants are used in more than one submodel. Classification in this table is made according to their first appearance in the model.

$$\frac{dD_{\text{P}}}{dt} = \frac{d_{\text{OT}}\Phi}{k_d}(\psi_{\text{OT}}^{\text{P}} - \Gamma) \quad \text{when } \Psi_{\text{OT}}^{\text{P}} > \Gamma, \quad \text{Eqn 12}$$

$$\frac{dD_{\text{P}}}{dt} = 0 \quad \text{when } \Psi_{\text{OT}}^{\text{P}} \leq \Gamma, \quad \text{Eqn 13}$$

where Φ (MPa⁻¹ h⁻¹) is the extensibility of cell walls in relation to nonreversible dimensional changes, and k_d (m m⁻¹) represents the contribution of changes in thickness of outer tissues (d_{OT}) relative to changes in total stem diameter (dD_{OT}/dD). Therefore:

$$\frac{dd_{\text{OT}}}{dt} = k_d \frac{dD}{dt}. \quad \text{Eqn 14}$$

Changes in d_{OT} are assumed to be proportional to those in stem diameter (D). Measurements with two point dendrometers located at the bark and the xylem showed that this linear fit optimised d_{OT} simulation (Supporting Information Fig. S1), previously described by a nonlinear fit following Génard *et al.* (2001). Note that for large trees, this relationship could deviate from a linear one and could be refined by analysing measurements with two point dendrometers as indicated above. The allometric nonlinear fit was uniquely applied to estimate initial d_{OT} (Fig. S2).

Elastic tension-driven changes in stem D (dD_{ET}/dt) are derived from Hooke's law and given by (cf. Eqn 24 in Steppe *et al.*, 2006):

$$\frac{dD_{\text{ET}}}{dt} = \frac{2d_{\text{OT}}}{\varepsilon} \frac{d\psi_{\text{OT}}^{\text{P}}}{dt}. \quad \text{Eqn 15}$$

Osmotically driven changes in water content (Mencuccini *et al.*, 2013) of outer tissues uniquely result from fluctuations in the concentration of carbohydrates because tension-driven changes are calculated independently. Hence, for a constant Ψ_{OT} ($d\Psi_{\text{OT}}/dt = 0$), any change in the osmotic component of Ψ_{OT} is compensated by an equal change of the pressure component in absolute terms (Eqn S13 in Mencuccini *et al.*, 2017). In this way, osmotically driven changes in stem D (dD_{EO}/dt) can be estimated applying Hooke's law (cf. Eqn S14 in Mencuccini *et al.*, 2017) likewise tension-driven changes (dD_{ET}/dt) are estimated:

$$\frac{dD_{\text{EO}}}{dt} = \frac{2d_{\text{OT}}}{\varepsilon} \frac{d\psi_{\text{OT}}^{\text{O}}}{dt}. \quad \text{Eqn 16}$$

Finally, the time derivative of D_{XY} is estimated according to the geometry of the stem:

$$\frac{dD_{XY}}{dt} = \frac{dD}{dt} - 2 \frac{dd_{OT}}{dt}. \quad \text{Eqn 17}$$

CO₂ sources and fluxes submodel ('submodel B') Growth and respiratory fluxes are expressed in moles of C and CO₂, respectively, and on a **stem volume basis** accounting for the relative portion of xylem and outer tissues. Once plastic variation in stem diameter (dD_P/dt) is separated from elastic fluctuations, stem growth (G , mol C m⁻³ h⁻¹) is calculated as the time derivative of new biomass added to the stem. For this, new biomass is estimated as the product of new volume by its density, and new volume is derived with respect to D_P assuming a cylindrical geometry of the stem (see Notes S1a for details):

$$G = \frac{D_P}{D^2} \frac{dD_P}{dt} (\rho_{OT} + \rho_{XY}) \frac{C_C}{M_C}, \quad \text{Eqn 18}$$

where ρ_{OT} and ρ_{XY} (g m⁻³) is the density of the outer tissues and xylem, respectively, C_C (g C g⁻¹) is stem C content and M_C (g mol⁻¹) is C molar mass. Note that the initial value of D_P is assumed equal to that of D_{XY} , and plastic growth accounts for cambial differentiation into both xylem and outer tissues cells (even if growth is assumed to uniquely occur in the outer tissues compartment to keep assumptions from submodel A valid). According to the GMRP, growth respiration (R_G , mol CO₂ m⁻³ h⁻¹) is estimated as a function of G and the growth yield (Y_G , mol C mol C⁻¹), defined as the ratio between new structural matter added to the stem and the amount of assimilates required for it (Amthor, 2000):

$$R_G = G \frac{1}{1 + Y_G}. \quad \text{Eqn 19}$$

Maintenance respiration (R_M , mol CO₂ m⁻³ h⁻¹) is sensitive to temperature (Amthor, 2000; Tjoelker *et al.*, 2001) and is estimated on a N content basis (Ryan, 1991; Reich *et al.*, 2008). Total R_M is allocated into xylem and outer tissues compartments ($R_M = R_{M_XY} + R_{M_OT}$) according to their corresponding N concentration ($[N]$, g N g⁻¹), and accounting for stem allometry (see Notes S1b for details):

$$R_{M_XY} = R_{b_N} [N]_{XY} \frac{\rho_{XY}}{M_N} Q_{10}^{\frac{T-T_b}{10}} \frac{V_{XY}}{V_S}, \quad \text{Eqn 20}$$

$$R_{M_OT} = R_{b_N} [N]_{OT} \frac{\rho_{OT}}{M_N} Q_{10}^{\frac{T-T_b}{10}} \frac{V_S - V_{XY}}{V_S}, \quad \text{Eqn 21}$$

where R_{b_N} (mol CO₂ mol N⁻¹ h⁻¹) is respiration at basal temperature (T_b , °C) on a N molar basis, M_N (g mol⁻¹) is the molar mass of N, Q_{10} is the temperature sensitivity of R_M , T is the stem temperature, and V_{XY} and V_S (m³) are the volume of the xylem and the stem within the monitored segment (considering the

length L_S (m) of the monitored segment). Average values of $[N]$ across cell tissues in both xylem and outer tissues are considered to represent R_M on a volume basis without distinction among cell tissues. Note that in this study case **there was no heartwood** (see 'Measurements and plant material') and the whole xylem compartment was assumed to respire. To apply TReSpire in mature trees with heartwood, a third coaxial cylinder should be added into the stem structure to account for stem allometry while differentiating between living/respoding compartments (xylem sapwood and outer tissues) and nonrespoding ones (heartwood).

Stem respiration (R_S , mol CO₂ m⁻³ h⁻¹) is the sum of the growth and maintenance components:

$$R_S = R_G + R_M = R_G + R_{M_XY} + R_{M_OT}. \quad \text{Eqn 22}$$

Once total R_S is known, CO₂ solubility in xylem sap and radial CO₂ diffusivity are calculated. According to **Henry's law**, the concentration of CO₂ dissolved in sap solution ($[CO_2^*]$, mol CO₂ L⁻¹) is proportional to the partial pressure of CO₂ in the gas phase in the xylem (pCO_2 , atm) (Hari *et al.*, 1991; Levy *et al.*, 1999; McGuire & Teskey, 2002; Hölttä & Kolari, 2009):

$$[CO_2^*] = k_H \left(1 + \frac{k_1}{10^{-pH}} + \frac{k_1 k_2}{(10^{-pH})^2} \right) pCO_2, \quad \text{Eqn 23}$$

where k_H (mol atm⁻¹ L⁻¹), k_1 and k_2 are temperature-dependent solubility coefficients for CO₂ in water (see Notes S1c for details), and pH is sap pH. Assuming ideal behaviour, xylem CO₂ concentration in the gas phase ($[CO_2]_{XY}$, mol CO₂ m⁻³) is:

$$[CO_2]_{XY} = \frac{pCO_2}{RT} = \frac{[CO_2^*]}{k_H \left(1 + \frac{k_1}{10^{-pH}} + \frac{k_1 k_2}{(10^{-pH})^2} \right) RT}, \quad \text{Eqn 24}$$

where R (m³ atm K⁻¹ mol⁻¹) is the ideal gas constant and T (K) is stem temperature. Note that the * superscript refers to the liquid phase to differentiate it from the gas phase.

The mass balance approach (MBA) proposed by McGuire & Teskey (2004) calculates R_S as the sum of stem CO₂ efflux, xylem CO₂ transport and the storage flux. Here, we refine the MBA by taking into account the geometry and the spatial location of different compartments proposed in TReSpire. To this end, two parallel mass balances in xylem and outer tissues are simultaneously applied to estimate R_S . Respired CO₂ in each compartment is the sum of the corresponding R_M (R_{M_XY} or R_{M_OT}) and half of R_G , assuming that CO₂ derived from the synthesis of new tissues in the cambium layer equally diffuses to the xylem and outer tissues following periclinal division of cambial cells. In this way, **half of R_G radially diffuses to the xylem** (Eqn 25), and the second half of R_G to the outer tissues (Eqn 26). Additionally, storage flux integrates CO₂ build-up in both gas and liquid phases (ΔS and ΔS^* , respectively) (cf. McGuire & Teskey, 2004). In this way, the mass balance in the xylem is given by:

$$R_{M_XY} + \frac{R_G}{2} = E_{XY} + F_T + (\Delta S_{XY} + \Delta S_{XY}^*), \quad \text{Eqn 25}$$

where E_{XY} ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) is CO_2 efflux from xylem to outer tissues, F_T ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) is axial CO_2 flux through the xylem, ΔS_{XY} ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) and ΔS_{XY}^* ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) is the storage flux in the xylem in gas and liquid phase, respectively. Similarly, the mass balance in outer tissues is given by:

$$R_{M_OT} + \frac{R_G}{2} = (E_A - E_{XY}) + \Delta S_{OT}, \quad \text{Eqn 26}$$

where E_A ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) is the CO_2 efflux from outer tissues to the atmosphere, and ΔS_{OT} ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) is the storage flux in outer tissues in the gas phase. Note that F_T and ΔS^* in outer tissues are neglected for model simplicity due to the small V_{OT} relative to V_S ($d_{OT} < 5\%$ of D ; Fig. S2), fraction directly related to the magnitude of these fluxes (see equations below).

Radial E_{XY} and E_A follows Fick's law of diffusion. According to the geometry of the xylem and outer tissues (see Notes S1d for details):

$$E_{XY} = \text{Dif} \frac{[\text{CO}_2]_{XY} - [\text{CO}_2]_{OT}}{\frac{2D_{XY}}{3\pi}} \frac{A_{XY}}{V_S}, \quad \text{Eqn 27}$$

$$E_A = \text{Dif} \frac{[\text{CO}_2]_{OT} - [\text{CO}_2]_{ATM}}{\frac{d_{OT}}{2}} \frac{A_S}{V_S}, \quad \text{Eqn 28}$$

where Dif ($\text{m}^2 \text{ h}^{-1}$) is diffusivity, $[\text{CO}_2]_{XY}$, $[\text{CO}_2]_{OT}$ and $[\text{CO}_2]_{ATM}$ are the gaseous $[\text{CO}_2]$ ($\text{mol CO}_2 \text{ m}^{-3}$) in xylem, outer tissues and atmosphere, respectively, and A_{XY} and A_S are the axial surface (m^2) of xylem and stem within the monitored stem segment, respectively. Given that CO_2 diffusion is $c. 10^4$ faster in air than in water (Nobel, 1999), diffusivity is exponentially related to the volumetric gas fraction (VGF) in the stem (VGF, $\text{m}^3 \text{ m}^{-3}$) (Sorz & Hietz, 2006):

$$\text{Dif} = \text{Dif}_0 e^{\text{Dif}_e \text{VGF}}, \quad \text{Eqn 29}$$

where Dif_0 ($\text{m}^2 \text{ h}^{-1}$) is diffusivity at zero VGF and Dif_e determines the shape of the exponential increase. Stem volume can be fractioned into gas, water and cell wall (VGF, VWF and VC_{WF} , respectively, $\text{m}^3 \text{ m}^{-3}$) (Gartner *et al.*, 2004). Assuming constant VC_{WF} , VGF is expected to increase to the detriment of VWF and *vice versa*:

$$\text{VGF} = (1 - \text{VWF} - \text{VC}_{WF}), \quad \text{Eqn 30}$$

and variation in VWF is assumed to be proportional to that in W_{OT} (from submodel A):

$$\text{VWF} = \text{VWF}_{t0} \left(1 - \frac{(W_{OT})_{t0} - W_{OT}}{(W_{OT})_{t0}} \right), \quad \text{Eqn 31}$$

where VWF_{t0} and $(W_{OT})_{t0}$ is the initial VWF and the initial W_{OT} , respectively. For model simplicity and to avoid over-parameterisation, VGF and Dif are assumed equal in xylem and outer tissues.

Xylem CO_2 transport (F_T) is calculated following McGuire & Teskey (2004):

$$F_T = \text{SF} \Delta[\text{CO}_2^*] \frac{1}{V_S}, \quad \text{Eqn 32}$$

where SF is sap flow ($\text{m}^3 \text{ h}^{-1}$) and $\Delta[\text{CO}_2^*]$ ($\text{mol CO}_2 \text{ m}^{-3}$) is the vertical gradient in $[\text{CO}_2^*]$ along the monitored stem segment. Note F_T is expressed on a stem volume basis (cf. McGuire & Teskey, 2004; see Notes S1e for details). Given that $\Delta[\text{CO}_2^*]$ is rarely constant and can be altered by for example transpiration and respiratory rates, it is allowed to vary with time in a linear manner:

$$\frac{d\Delta[\text{CO}_2^*]}{dt} = \Delta[\text{CO}_2^*]_{\text{slope}}, \quad \text{Eqn 33}$$

where $\Delta[\text{CO}_2^*]_{\text{slope}}$ ($\text{mol CO}_2 \text{ m}^{-3} \text{ h}^{-1}$) represents the slope of the variation of $\Delta[\text{CO}_2^*]$ with time.

To close the mass balance in outer tissues, ΔS_{OT} is calculated once all other terms in Eqn 26 are known. The time derivative of $[\text{CO}_2]_{OT}$ is estimated, by definition (McGuire & Teskey, 2004), as follows:

$$\frac{d[\text{CO}_2]_{OT}}{dt} = \Delta S_{OT} \frac{1}{\text{VGF}}. \quad \text{Eqn 34}$$

Similarly, ΔS_{XY} is estimated as a function of the time derivative of $[\text{CO}_2]_{XY}$:

$$\Delta S_{XY} = \frac{d[\text{CO}_2]_{XY}}{dt} \text{VGF}. \quad \text{Eqn 35}$$

Finally, ΔS_{XY}^* is calculated once all other terms in Eqn 25 terms are known, and it equals the time derivative of $[\text{CO}_2^*]$ accounting for the amount of water in the stem segment:

$$\frac{d[\text{CO}_2^*]}{dt} = \Delta S_{XY}^* \frac{1}{\text{VWF} \rho_w}, \quad \text{Eqn 36}$$

where ρ_w is water density.

Stem carbon balance ('submodel C') Phloem unloading, sugar-starch interconversion, growth, respiration and osmoregulation, expressed in sucrose equivalents and on a stem volume basis ($\text{g sucrose h}^{-1} \text{ m}^{-3}$) are quantified to close the stem C balance (Lacointe & Minchin, 2008; De Schepper & Steppe, 2010).

Phloem unloading (U) is derived from the vertical gradient in turgor pressure of phloem cells along the stem by means of monitoring stem D variations (ΔD) across the stem length (Sevanto *et al.*, 2014) (i.e. at the stem base and at the upper stem just below the first branch). The first part of submodel A

(Eqns 1–6, the so-called ‘water transport submodel’ in Steppe *et al.* (2006, 2008)), is assumed to behave equally along the vertical gradient, and the second part of submodel A (Eqns 7–17, the so-called ‘stem diameter variation submodel’ in Steppe *et al.* (2006, 2008)) is calibrated and run for a second time according to ΔD in the upper part of the stem (D_{UP}). Note that submodel A is calibrated against stem ΔD at the stem base (see ‘Measurements and plant material’ section), and to avoid model over-parameterisation, parameters applied and calibrated for the lower part of the stem are assumed to remain constant along the vertical gradient. Likewise, measured and simulated variables included in Eqns 1–6 determining axial and radial water flows (SF, f_{OT} , Ψ_{XY} and Ψ_{OT}) are assumed to be the same in the lower and upper stem, and variables from which D_{UP} is estimated (Eqns 7–17; ϵ_{UP} , V_{OT_UP} , $\Psi_{OT_UP}^O$, $\Psi_{OT_UP}^P$, D_{XY_UP} , d_{OT_UP} , D_{P_UP} , dD_{ET_UP} and dD_{EO_UP}) are simulated for a second time for the upper part of the stem. In this way, the osmotic and pressure components of Ψ_{OT} do change with stem height, and the pressure drop along the stem ($\Delta\Psi_{OT}^P$, MPa) is given by the difference between Ψ_{OT}^P (MPa) in the upper and lower stem accounting for the gravitational pressure difference (Hölttä *et al.*, 2009):

$$\Delta\Psi_{OT}^P = (\Psi_{OT_UP}^P - \Psi_{OT}^P) + (\rho_w g L), \quad \text{Eqn 37}$$

where g (MPa m² g⁻¹) is the gravitational constant. Note that the subscript for the lower part of the stem is omitted for consistency with the terminology proposed in submodel A. According to Münch theory, variations in the pressure component of Ψ_{OT} along the stem are driven by variations in the osmotic component due to loading-unloading phloem dynamics (De Schepper & Steppe, 2010). Therefore, for a given Ψ_{OT} , the vertical drop in Ψ_{OT}^P equals that of Ψ_{OT}^O ($\Delta\Psi_{OT}^P = \Delta\Psi_{OT}^O$), and the vertical gradient in sucrose concentration in outer tissues along the stem ($\Delta[SU]_{OT}$, g m⁻³) can be estimated with the van ’t Hoff equation (see Notes S1f for details):

$$\Delta[SU]_{OT} = \frac{\Delta\Psi_{OT}^P M_{SU}}{RT}, \quad \text{Eqn 38}$$

where M_{SU} (g mol⁻¹) is the molar mass of sucrose, R (m³ MPa K⁻¹ mol⁻¹) is the universal gas constant and T (K) is temperature. Single phase Michaelis–Menten kinetics describe U (g m⁻³ h⁻¹) rates (Minchin & Lacombe, 2017) (see Notes S1g for details):

$$U = \frac{V_{\max} M_{SU} \Delta[SU]_{OT}}{k_M M_{SU} + \Delta[SU]_{OT}}, \quad \text{Eqn 39}$$

where $V_{\max}$ (mol m⁻³ h⁻¹) and k_M (mol m⁻³) are kinetic parameters.

The time derivative of starch concentration ([ST], g m⁻³) is determined by a constant target concentration of sucrose ([SU]_{targ}, g m⁻³) in the outer tissues (Lacombe & Minchin, 2008; De Schepper & Steppe, 2010):

$$\frac{d[ST]}{dt} = k_{\text{starch}} ([SU] - [SU]_{\text{targ}}), \quad \text{Eqn 40}$$

where [SU] is the sucrose concentration in outer tissues and k_{starch} (h⁻¹) is a kinetic parameter. When [SU] (g m⁻³) is higher than [SU]_{targ}, starch is synthesised and *vice versa*, so that starch synthesis and depolymerisation are given by positive and negative values, respectively. For calibration purposes, starch storage is assumed to occur in the xylem. Sugar–starch interconversion (SSI, g m⁻³ h⁻¹) is therefore given by:

$$SSI = \frac{d[ST]}{dt} \frac{V_{XY}}{V_S}. \quad \text{Eqn 41}$$

Stem growth and respiration sinks (G_{SU} and R_{S_SU} , g m⁻³ h⁻¹, respectively; in sucrose equivalents) are quantified as follows:

$$G_{SU} = G \frac{M_{SU}}{12}, \quad \text{Eqn 42}$$

$$R_{S_SU} = R_S \frac{M_{SU}}{12}, \quad \text{Eqn 43}$$

with 12 being the ratio of C (or CO₂) mol to sucrose (C₁₂H₂₂O₁₁) mol.

In this case study, woody tissue photosynthesis was disabled (see ‘Measurements and plant material’ section) and hence neglected for modelling purposes. Finally, the time derivative of [SU] in outer tissues is given by:

$$\frac{d[SU]}{dt} = (U - G_{SU} - R_{S_SU} - SSI) \frac{V_S}{V_S - V_{XY}}, \quad \text{Eqn 44}$$

where $V_S/(V_S - V_{XY})$ accounts for stem allometry assuming that osmoregulation ($OR = d[SU]/dt$, g m⁻³ h⁻¹) uniquely occurs in

Table 3 Summary of variables needed to run and calibrate the model.

Variable	Purpose ¹	Submodel	Method ²
Soil water potential (Ψ_{SOIL})	Input	A	Tensiometer
Sap flow (SF)	Input	A	Heat balance
Stem diameter variations (ΔD)	Calibration	A	LVDT ³
Xylem water potential (Ψ_{XY})	Calibration	A	Pressure chamber
Stem temperature (T)	Input	B	Thermocouple
Stem CO ₂ efflux (E_A)	Calibration	B	IRGA
Upper stem ΔD (ΔD_{UP})	Calibration	C	LVDT ³
Soluble sugars concentration ([SU])	Calibration	C	Spectrophotometry
Starch concentration ([ST])	Calibration	C	Spectrophotometry

¹Input variables need to be continuously monitored, while calibration variables can be discretely measured.

²Method applied in this study case.

³Linear Variable Displacement Transducer. An additional LVDT placed at the xylem surface would be necessary if k_d is not derived from measurements in ancillary trees.

the outer tissues compartment given that the osmotic component of Ψ_{XY} is zero (Steppe *et al.*, 2006).

Measurements and plant material

To run and calibrate TReSpire, an array of variables should be measured (Table 3). Briefly, SF and soil water potential are used as inputs in submodel A, which is calibrated against stem ΔD and xylem water potential. Submodel B is driven by previous model outputs and stem temperature, and is calibrated against E_A . Submodel C in turn is driven by previous model outputs and calibrated against stem ΔD at the upper part of the stem and concentrations of soluble sugars and starch. Additionally, some stem features included in TReSpire as parameters were measured or estimated from ancillary trees. These include L , initial D , k_d , ρ_{XY} , ρ_{OT} , $[N]$ and $[C]$ in xylem and outer tissues, and sap pH (Table S1). See Methods S1 for a detailed explanation of sensors and methods applied to measure abovementioned variables and parameters, and Methods S2 for a picture of the experiment setup.

To present TReSpire, data from a monitored Norway maple tree (*Acer platanoides* L. ‘Crimson King’) during the growing season in spring (DOYs 155–160) is shown. Data were collected in 2017 within the context of a larger experiment to evaluate TReSpire across different species, phenological stages and under drought stress conditions. In this larger experiment, ancillary trees of the same species, age, size, and grown under identical conditions were instrumented to monitor xylem $[CO_2]$ and xylem ΔD with intrusive methods to be avoided in modelling-dedicated trees. Data from ancillary trees is also presented here for model validation. Trees were grown outdoors in a plant nursery and transported to the glasshouse facilities of Ghent

University before measurements. Maple trees averaged 4 cm in diameter at 1 m height, 315 cm in total height and 180 cm of stem length before the first branch.

Model implementation and analyses

Model simulation, calibration, sensitivity and identifiability analyses were performed using the plant modelling software PHYTO-SIM (v.2.1; Phyto-IT, Gent, Belgium). Simulations started at nighttime before transpiration resumes, assuming hydraulic equilibrium along the root-to-leaf continuum. Simulations were performed with a fourth-order variable step size solver with an accuracy of 10^{-6} and a maximum step size of 0.1 h. Model parameterisation was performed by applying the Simplex optimisation method (Nelder & Mead, 1965) in order to minimise the weighted sum of squared errors between data and model simulation. Before model parameterisation, an identifiability analysis was performed to select an identifiable parameter subset following the procedure described in De Pauw *et al.* (2008). In order for a parameter subset to be identifiable from the available data, two conditions need to be met: (1) the subset needs to be as sensitive as possible; and (2) the correlations between the parameters of the subset need to be as low as possible (low collinearity). The remaining unidentifiable parameters were assigned values based on literature or previous modelling exercises (see Table S1 for further details). For the estimated parameters, 95% confidence intervals and relative standard errors were calculated based on the parameter estimation covariance matrix (Steppe *et al.*, 2006). The parameter estimation covariance matrix was calculated using a two-step procedure. First, it was calculated for the estimated parameters of submodels A and B, then it was calculated for the parameters of

Table 4 Parameters and initial values of derivative variables of submodels A and B (mean and 95% confidence interval), and their relative standard error (RSE) and sensitivity indexes against each calibration variable: xylem water potential, stem diameter and stem CO_2 efflux to the atmosphere (Ψ_{XY} , D and E_A , respectively).

Submodel & Parameter		Confidence analysis		Sensitivity analysis		
		Mean [95% CI]	RSE (%)	Ψ_{XY}	D	E_A
A	R_{XY}	1.03×10^{-3} [1.02×10^{-3} , 1.03×10^{-3}]	0.30	198.15	23.98	0.26
	R_{OT}	3.43×10^{-1} [3.35×10^{-1} , 3.51×10^{-1}]	1.17	43.32	3.33	0.13
	C_{OT}	$1.31 \times 10^{+1}$ [$1.31 \times 10^{+1}$, $1.32 \times 10^{+1}$]	0.35	0.94	31.66	0.30
	ε_0	NA		52.33	23.90	0.19
	Φ	9.34×10^{-4} [9.30×10^{-4} , 9.38×10^{-4}]	0.22	32.19	74.19	0.93
B	Y_G	8.18×10^{-1} [8.12×10^{-1} , 8.24×10^{-1}]	0.40	46.81	1.64	4.66
	$R_{b,N}$	9.95×10^{-3} [9.63×10^{-3} , 1.03×10^{-2}]	1.63	1.47	0.02	1.75
	Q_{10}	$2.18 \times 10^{+0}$ [$2.10 \times 10^{+0}$, $2.27 \times 10^{+0}$]	1.93	4.11	0.14	1.01
	Dif_e	$2.93 \times 10^{+1}$ [$2.91 \times 10^{+1}$, $2.94 \times 10^{+1}$]	0.25	32.52	1.13	7.32
	Dif_o	NA		17.38	0.60	0.82
	$\Delta[CO_2^*]_{slope}$	-5.84×10^{-3} [-5.96×10^{-3} , -5.71×10^{-3}]	1.10	9.11	0.31	0.25
	$(\Delta[CO_2^*])_{t0}$	7.20×10^{-1} [6.96×10^{-1} , 7.44×10^{-1}]	1.68	7.12	0.25	0.50
	$([CO_2^*])_{t0}$	5.50×10^{-3} [5.32×10^{-3} , 5.67×10^{-3}]	1.64	17.70	0.63	0.14
	$([CO_2]_{OT})_{t0}$	5.34×10^{-1} [5.18×10^{-1} , 5.51×10^{-1}]	1.58	10.27	0.35	0.83

NA denote unidentifiable parameters. $t0$ subscript denote the initial value of derived variables. For calibration and sensitivity analyses, scaling values of measured variables Ψ_{XY} , D and E_A were 0.002 MPa, 3×10^{-6} m and $0.5 \text{ mol } CO_2 \text{ m}^{-3} \text{ h}^{-1}$, respectively, according to measurement error standard deviation. Due to the relatively low frequency of Ψ_{XY} measurements, its scaling value was deliberately reduced to increase its objective weight, so that calibration was not biased towards D and E_A . See Tables 1 and 2 for the definition of the parameter acronyms.

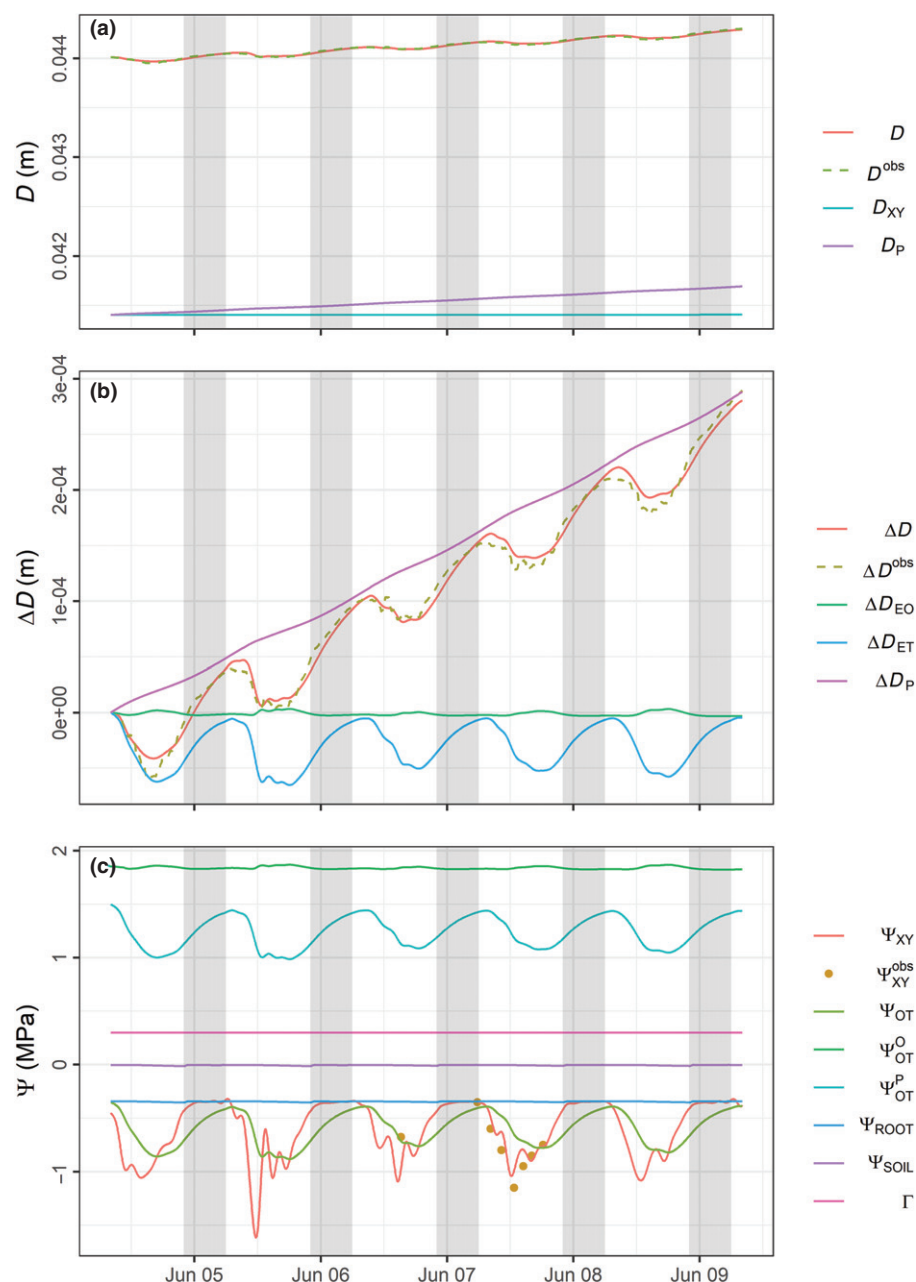


Fig. 2 Water flow and storage submodel. (a) Stem diameter measurements (D^{obs}) and simulation (D) with xylem diameter (D_{XY}) and plastic diameter (D_P). (b) Contribution of plastic irreversible growth (ΔD_P), elastic osmotically driven changes (ΔD_{EO}) and elastic tension-driven changes (ΔD_{ET}) to total stem D fluctuations (ΔD). Measured stem D fluctuations (ΔD^{obs}) are additionally plotted for comparison with simulation. (c) Measurements of xylem water potential (Ψ_{XY}^{obs}) and simulation of xylem (Ψ_{XY}) and outer tissues (Ψ_{OT}) water potential. The latter results from the sum of osmotic (Ψ_{OT}^O) and turgor pressure (Ψ_{OT}^P) components. Soil (Ψ_{SOIL}) and root (Ψ_{ROOT}) water potential are also displayed together with the threshold value for irreversible growth (Γ). Shaded areas indicate nighttime.

submodel C. This was done to avoid numerical errors which arose when calculating the matrix for the entire estimated parameter subset. The poor identifiability of parameters in submodel C caused the entire matrix to become near-singular, which is known to cause problems when trying to invert it. Although we were not able to obtain a parameter estimation error for submodel C, the estimated values itself were realistic and did not cause problems when running the model.

Data availability

The model (PHYTOSIM workspace) and the corresponding data that support the findings of this study are openly available in 'Mendeley data' at doi: 10.17632/9c9w7mv9y9d.1.

Results

Table 4 shows the largest subset of identifiable parameters of submodels A and B with highest sensitivity index (R_{XY} , R_{OT} , C_{OT} , Φ , Y_G , R_{b_N} , Q_{10} , Dif_e and $\Delta[CO_2^*]_{slope}$), which was calibrated with a relative standard error $\leq 2\%$. Due to a high collinearity between Dif_e and Dif_0 and between C_{OT} and ϵ_0 , parameters Dif_0 and ϵ_0 were not included in the calibration and values from literature or preliminary calibrations were applied (Table S1). Initial values of derivative variables $[CO_2^*]$, $[CO_2^*]_{OT}$ and $\Delta[CO_2^*]$ were also identifiable. Ψ_{XY} was highly sensitive to R_{XY} , stem D to Φ , and E_A to both Dif_e and Y_G , followed by R_{b_N} and Q_{10} . Some parameter subsets from submodel C showed high collinearities (e.g. between V_{max} and K_M , and between k_{starch} and $[SU]_{targ}$).

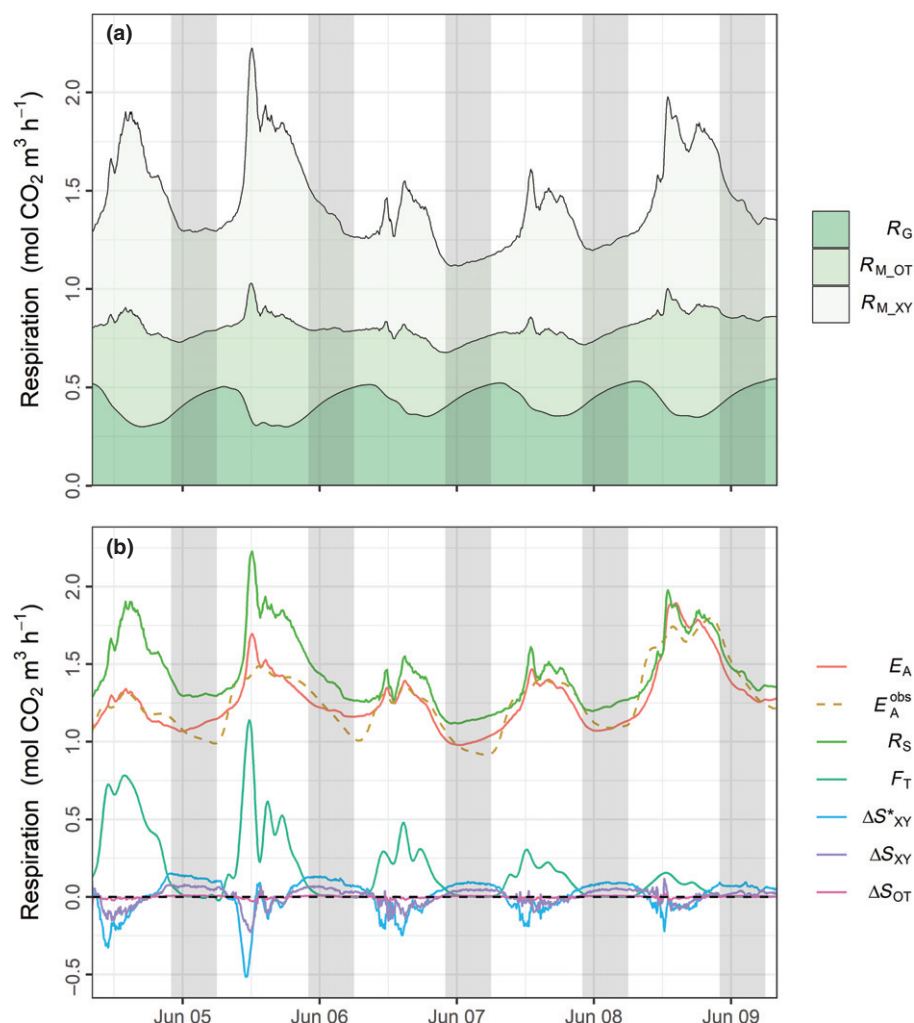


Fig. 3 CO₂ sources and fluxes submodel. (a) Partitioning of stem respiration into growth respiration (R_G), maintenance respiration in outer tissues (R_{M_OT}) and maintenance respiration in xylem (R_{M_XY}). (b) Stem respiration (R_S) is the sum of stem CO₂ efflux to the atmosphere (E_A), xylem CO₂ transport through xylem (F_T) and the storage flux, which occurs in the xylem in liquid (ΔS^*_{XY}) and gas (ΔS_{XY}) phases and in outer tissues in the gas phase (ΔS_{OT}). Simulations are calibrated against measurements of E_A (E_A^{obs}). Shaded areas indicate nighttime.

Therefore, only the subset of parameters constituted by V_{max} and k_{starch} and the initial value of $\Psi_{OT\ UP}^P$ were calibrated (Table S1), while K_M , $[SU]_{targ}$ and the initial values of $[SU]$ and $[ST]$ were taken from literature or using expert knowledge (Table S1). See Table S2 for the sensitivity analysis of submodel C.

Submodel A accurately simulated stem D variations and Ψ_{XY} (Fig. 2). Three sources of fluctuations in stem D were disentangled. Driven by turgor pressure, irreversible plastic growth (dD_P/dt) occurred throughout the subdaily period with higher rates at nighttime. Elastic tension-driven changes (dD_{ET}/dt) caused substantial daytime shrinkage and nighttime swelling, while osmotically driven changes (dD_{EO}/dt) were comparatively much smaller and exhibited the opposite pattern (Fig. 2a,b). Ψ_{OT} lagged behind Ψ_{XY} and its pressure component fluctuated more than the osmotic counterpart (Fig. 2c).

Stem respiration fluctuated between 1.1 and 2.2 mol CO₂ m⁻³ h⁻¹ and was partitioned into R_G , R_{M_OT} and R_{M_XY} (Fig. 3a). R_G mimicked dD_P/dt subdaily dynamics, with highest rates at nighttime up to 0.5 mol CO₂ m⁻³ h⁻¹. Conversely, R_M in xylem and outer tissues were temperature dependent and exhibited highest rates during daytime, with R_{M_XY} being higher than R_{M_OT} . Simulated E_A fluctuated between 1.0 and

1.9 mol CO₂ m⁻³ h⁻¹, accounting for most of R_S (Fig. 3b). Daytime E_A was accurately simulated, whereas nighttime E_A was slightly overestimated. F_T was the second largest component of R_S and was driven by SF dynamics. F_T reached an initial peak of 1.1 mol CO₂ m⁻³ h⁻¹, and progressively decreased to a subdaily maximum of 0.2 mol CO₂ m⁻³ h⁻¹ following a reduction in the vertical gradient of sap $[CO_2^*]$ across the surveyed period ($\Delta [CO_2^*]_{slope} = -5.8 \times 10^{-3}$ mol CO₂ m⁻³ h⁻¹). Storage fluxes were the smallest component of R_S , with xylem storage in both liquid (ΔS^*_{XY}) and gaseous (ΔS_{XY}) phase accounting for most of it, while CO₂ storage in outer tissues (ΔS_{OT}) was negligible. CO₂ was stored at nighttime and released at daytime.

Submodel B yielded additional insights into R_S and related respiratory fluxes (Fig. 4). Maintenance respiration (integrating xylem and outer tissues) roughly accounted for 55–85% of total R_S , with R_{M_XY} contributing with 35–55% to R_S . E_A accounted for 85–95% of R_S at nighttime, with minimum values down to 70% during daytime (Fig. 4a). Xylem CO₂ concentration in the gas and liquid phase ranged between 2.0 and 4.9 mol m⁻³, and between 2.5 and 6.8 mM, respectively (Fig. 4b); values consistent with measurements performed in ancillary trees (Fig. S3a,c). In outer tissues, $[CO_2]$ was substantially lower (0.3–0.6 mol m⁻³).

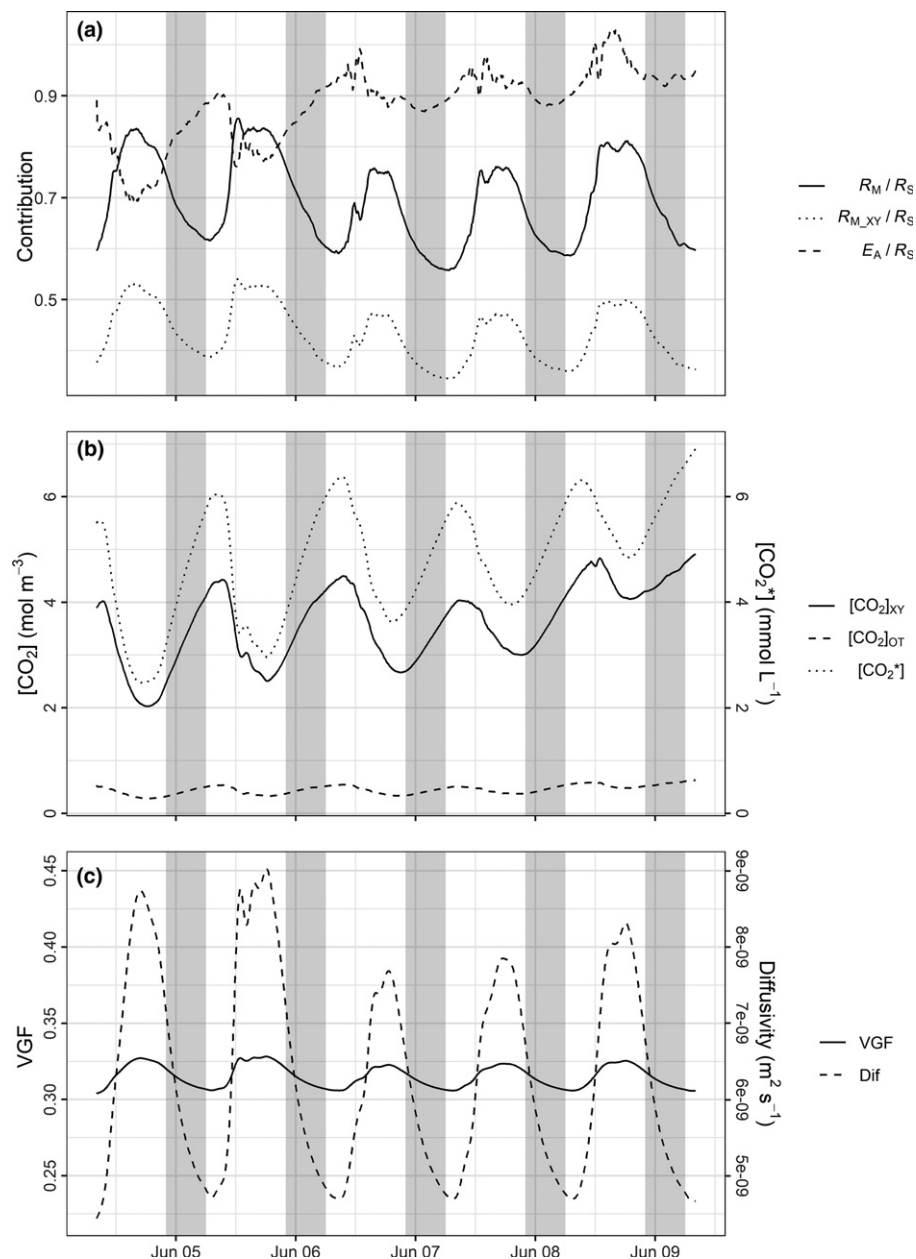


Fig. 4 Additional outputs of the CO₂ sources and fluxes submodel. (a) Contribution of maintenance respiration of xylem and outer tissues to stem respiration (R_M/R_S), xylem maintenance respiration to stem respiration (R_{M_XY}/R_S), and the ratio of stem CO₂ efflux to stem respiration (E_A/R_S). (b) Concentration of gaseous CO₂ in the xylem ($[CO_2]_{XY}$) and outer tissues ($[CO_2]_{OT}$), and concentration of CO₂ dissolved in the sap solution ($[CO_2]^*$). (c) Subdaily variation in volume gas fraction (VGF) in the stem and corresponding radial CO₂ diffusivity (Dif). Shaded areas indicate nighttime.

Subdaily dynamics in $[CO_2]^*$, $[CO_2]_{XY}$ and $[CO_2]_{OT}$ were opposite to those of stem temperature (cf. Fig. S3) according to Henry's and ideal gas laws. Radial CO₂ diffusivity fluctuated between 5×10^{-9} at nighttime and $9 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ at daytime (Fig. 4c), which was driven by transpiration dynamics and the resulting volumetric gas fraction.

The stem C balance (submodel C) was closed by assigning values from literature to unidentifiable parameters (Table S1). Model fit of stem ΔD in the upper part stem (Fig. 5a) was worse than that at the stem base (cf. Fig. 2a). Both Ψ_{OT}^P and Ψ_{OT}^O were higher (in absolute terms) in the upper stem resulting in a vertical pressure gradient fluctuating between 0.1 and 0.2 MPa (Fig. 5b). Across the surveyed period, [SU] was roughly stable and below $[SU]_{\text{targ}}$ resulting in a progressive reduction of [ST] by means of starch depolymerisation (Fig. 5c). Phloem unloading (76–

89 $\text{g m}^{-3} \text{ h}^{-1}$) and starch depolymerisation (10–11 $\text{g m}^{-3} \text{ h}^{-1}$) fuelled growth (38–70 $\text{g m}^{-3} \text{ h}^{-1}$) and respiration (32–63 $\text{g m}^{-3} \text{ h}^{-1}$). Mismatch between C sources and sinks were compensated for by osmoregulation (defined as the time derivative of [SU]), which fluctuated between -16 and $11 \text{ g m}^{-3} \text{ h}^{-1}$ (Fig. 5d), therefore acting as a C sink or source according to C metabolic requirements.

Discussion

Submodels A and B were calibrated with high level of statistical confidence (Table 4) and were useful to qualitatively and quantitatively describe stem respiration. Submodel A was pivotal to simulate turgor pressure and separate plastic stem growth from elastic fluctuations, and subsequently discriminate between

growth and maintenance respiration on a subdaily basis. In this way, TReSpire was calibrated against stem diameter variations and stem CO₂ efflux, and hence from a sink-driven perspective (Fatichi *et al.*, 2014, 2019; Steppe *et al.*, 2015). Submodel B was useful to determine key parameters described by the GMRP to estimate whole-plant respiration (Y_G , Q_{10} and R_{b_N}), and realistically simulated dynamics of CO₂ diffusion across the stem and CO₂ solubility in the sap solution, a crucial step to deepen our understanding of stem CO₂ fluxes (Teskey *et al.*, 2008, 2017; Trumbore *et al.*, 2013). Conversely, submodel C was practical to conceptually describe C sources and sinks at the organ level; however, refinement would be necessary before gained insights from this submodel are extrapolated to large-scale models, as collinearity issues could lead to misestimation of phloem unloading and SSI rates. In any case, given that stem hydraulic modelling (submodel A) has been explored and discussed to a great detail in previous works (Steppe *et al.*, 2006, 2015; Hölttä *et al.*, 2009; Mencuccini *et al.*, 2017), discussion here will deliberately focus on stem respiratory physiology (submodel B) and the stem carbon balance (submodel C), which encompass the largest advance in plant modelling proposed in TReSpire.

Insights into the growth and maintenance respiration paradigm

The GMRP separates respiration into growth and maintenance components. The growth component is directly related to the growth yield (Y_G), a parameter defined by the fraction of C added to structure per unit of substrate invested in growth metabolism. This parameter indicates how expensive is the synthesis of new biomass, and is therefore related to the species ecology (Lambers & Poorter, 1992; Rodríguez-Calcerrada *et al.*, 2019). Here, Y_G at the onset of the growing period (DOYs 155–160) was 0.818 ± 0.006 (95% CI), a coefficient at the upper end of reported values (0.7–0.85; Cannell & Thornley, 2000; Amthor, 2000). This relatively high Y_G could be attributed to seasonality in growth efficiency (Steppe *et al.*, 2015). Carbon investment in cell production and enlargement peaks at the beginning of the growing season, while C investment in cell wall thickening and lignification lags behind by 25–50 d depending on the forest biome (Cuny *et al.*, 2015). Given that lignification is comparatively more C demanding, Y_G might decline progressively as the contribution of lignification to stem size increment becomes more important. It is worth noting, however, that literature comparison is not straightforward as different growth-related subprocesses taken into account (e.g. N reduction and uptake, and phloem transport) or analytical methods applied for calculation, substantially influence Y_G estimates (Cannell & Thornley, 2000; Rodríguez-Calcerrada *et al.*, 2019).

Maintenance respiration relies on two parameters (R_{b_N} and Q_{10}). Basal respiration on a N basis at 20°C (R_{b_N}) was $9.95 \times 10^{-3} \pm 0.32 \times 10^{-3}$ mol CO₂ mol N⁻¹ h⁻¹, a value strikingly close to that reported by Ryan (1991) in a pioneering work (10.6×10^{-3} mol CO₂ mol N⁻¹ h⁻¹) that inspired plant respiration modellers. Such consistency highlights once more the tight link between [N] and the amount of proteins involved in

maintenance metabolism. We initially attempted to describe the R_{M-N} relationship with a power function following Reich *et al.*'s (2008). However, the intercept and slope of this log-log relationship were highly correlated and did not reduce the weighted sum of squared errors (data not shown), so that we adopted a linear approach. We suggest that the comparatively narrow range of [N] measured here in xylem and outer tissues leads to a R_{M-N} relationship that can be similarly described by linear and exponential shapes. Furthermore, when the R_{M-N} relationship was exponentially described, Y_G tended to the unit and hence $R_G \approx 0$ and $R_S \approx R_M$. Given that in Reich *et al.*'s (2008) meta-analyses, respiration rates pooled R_G and R_M , we speculate that R_{M-N} could be linearly described, while differences between exponential and linear predictions (R_G) could be attributed to additional N required to meet growth stoichiometry. This hypothesis, however, should be better approached with dedicated experiments.

On the other hand, temperature sensitivity of R_M (Q_{10}) was 2.18 ± 0.08 . This Q_{10} is within the reported range in stems (from 1 to 3, reviewed by Wang *et al.*, 2006 and Salomón *et al.*, 2017), moderately higher than the median (1.91; Wang *et al.*, 2006) and the Q_{10} value commonly used in plant and global models (2.0). Again, literature comparison is not straightforward as: Q_{10} decreases across the season as plants acclimate to warmer conditions (Tjoelker *et al.*, 2001; Reich *et al.*, 2016), and Q_{10} estimates can be biased by implicit hypotheses assumed in each study case. Many studies determine Q_{10} through the exponential relationship between R_S ($R_G + R_M$) and temperature. However, R_G is temperature independent *per se*, and uniquely indirect temperature effects on the sink demand (growth) affect associated respiratory costs (Amthor, 2000). Therefore, inclusion of R_G to estimate Q_{10} might blur the R_{M-T} relationship (Salomón *et al.*, 2018).

Insights into stem CO₂ fluxes

TReSpire applies a refined version of the MBA (McGuire & Teskey, 2004) including a parallel mass balance in the outer tissues in addition to that in the xylem. The spatial compartmentalisation of TReSpire allows us to discern the origin of respired CO₂ among xylem, outer tissues and the cambium layer, while intrusive measurement techniques are avoided. Overall, the relative contribution of E_A , F_T and ΔS to R_S , gaseous [CO₂] in xylem and outer tissues, and [CO₂*] dissolved in sap solution are consistent with observations made across a wide array of species and environmental conditions (reviewed by Teskey *et al.*, 2008, 2017), as well as with xylem [CO₂] measurements performed in ancillary trees instrumented for model validation (Fig. S3).

Radial CO₂ movement was driven by the [CO₂] gradient across the xylem-outer tissues-atmosphere continuum according to Fick's law of diffusion. Simulated xylem [CO₂] was approximately one and two orders of magnitude higher than that in outer tissues and the atmosphere, respectively, which is consistent with [CO₂] measurements along this gradient (reviewed by Teskey *et al.*, 2008). This gradient responds to slow radial CO₂ diffusion in woody tissues (Sorz & Hietz, 2006; Steppe *et al.*, 2007) and stem geometry. While outer tissues consist of a relatively thin superposition of cell layers peripherally located and

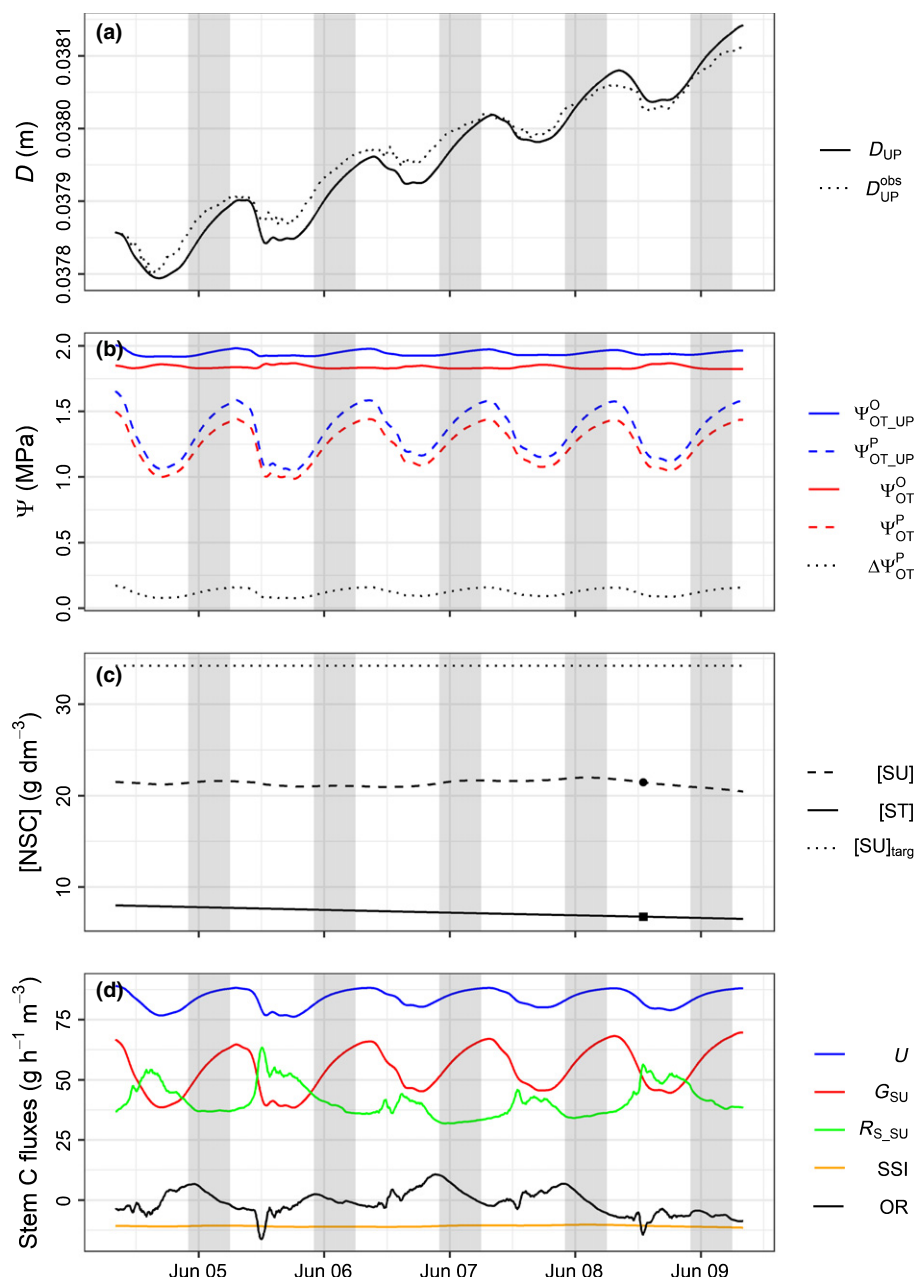


Fig. 5 Stem carbon balance submodel. (a) Stem diameter measurements (D_{UP}^{obs}) and simulation (D_{UP}) in the upper part of the stem. (b) Osmotic and pressure components of outer tissues water potential in the upper ($\Psi_{OT_UP}^O$, $\Psi_{OT_UP}^P$) and lower (Ψ_{OT}^O , Ψ_{OT}^P) part of the stem, and the resulting pressure drop along the stem ($\Delta\Psi_{OT}^P$). (c) Concentration of sucrose equivalents ([SU]), starch ([ST]) and target [SU] for starch metabolism ($[SU]_{targ}$). Punctual measurements of [SU] and [ST] used for calibration are shown by the closed circle and square, respectively, whereas [SU] and [ST] simulations are shown by dashed and continuous lines, respectively. NSC, nonstructural carbohydrates. (d) Stem carbon sources and sinks: phloem unload (U), stem growth and respiration in sucrose equivalents (G_{SU} and R_{S_SU} , respectively), sugar–starch interconversion (SSI) and osmoregulation (OR). Shaded areas indicate nighttime.

immediately diffusing to the atmosphere, CO_2 in the thicker xylem requires more time to reach the cambium layer facilitating CO_2 build-up (Eqns 27, 28). The high water status of our well watered monitored tree, resulted in a radial CO_2 diffusivity in woody tissues ($5\text{--}9 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$) higher than that in water ($1.6 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$), and three orders of magnitude lower than that in air ($1.6 \times 10^{-5} \text{ m}^2 \text{ s}^{-1}$), highlighting the substantial constraint of water and cell walls to CO_2 diffusion through woody tissues. Accordingly, our limited subdaily range of variation in diffusivity (cf. Sorz & Hietz, 2006) followed dynamics of volumetric gas fraction, as similarly observed in poplar trees under controlled conditions (Steppe *et al.*, 2007), and in field-grown oaks experiencing seasonal drought (Salomón *et al.*, 2016). To avoid model overparameterisation, TReSpire assumes equal diffusivity and volumetric water fraction between compartments,

which is unlikely due to differential structure and functionality of xylem and outer tissues (Steppe *et al.*, 2015). Measurements relating water content, wood anatomy and CO_2 diffusivity (Sorz & Hietz, 2006) would help to more realistically parameterise and model CO_2 diffusion through woody tissues.

Axial CO_2 transport through xylem (F_T) explained to the greatest extent the difference between R_S and E_A (Fig. 3b) and hence E_A/R_S deviation from unity (Fig. 4a). The stem apparent respiratory quotient ($ARQ = CO_2 \text{ efflux}/O_2 \text{ influx}$) is theoretically equivalent to the $E_A : R_S$ ratio (cf. Hölttä & Kolari, 2009) assuming that stem oxygen consumption equals stem respiration. Ratios of E_A/R_S ranging between 0.7 and 1 agree with ARQ s below one, consistently observed across a range of species and conditions (Angert *et al.*, 2012; Hilman *et al.*, 2018). However, subdaily variation in E_A/R_S according to transpiration dynamics

contrasts with mostly stable ARQs along the subdaily course (Hilman *et al.*, 2018). Comparable ARQs between day and night suggests that F_T is not the only path of 'CO₂ removal', and authors suggested that enzyme phosphoenolpyruvate carboxylase (PEPC) can fix locally respired CO₂ and export the resulting product via xylem and phloem flows. In this line, E_A was slightly overestimated at nighttime. TReSpire assumes that F_T is restricted to daytime hours following SF dynamics (Eqn 32). Hence, for a given R_S , reductions in E_A during nighttime can be uniquely achieved by increased CO₂ storage. However, ΔS is limited at some point by CO₂ saturation in sap according to equilibrium reactions, which are pH-dependent (Teskey *et al.*, 2008). F_T simulated here could be considered as a lumped flow able to remove locally respired CO₂ integrating CO₂ dissolved in sap (McGuire & Teskey, 2004), malate dissolved in sap after PEPC activity (Angert *et al.*, 2012; Hilman *et al.*, 2018), and axial CO₂ diffusion in the gas phase following vertical [CO₂] gradients resulting from woody tissue photosynthesis above and below the monitored stem segment (Saveyn *et al.*, 2008; De Roo *et al.*, 2019; Salomón *et al.*, 2019a). Whatever the pathway of CO₂ removal, F_T is restricted to daytime and further research on the role of PEPC activity would be necessary to integrate additional mechanisms of CO₂ removal at nighttime, as well as to address potential differences in subdaily patterns between E_A/R_S simulations and ARQs measurements. Furthermore, TReSpire allows temporal variation in the vertical gradient of sap [CO₂*] (Δ [CO₂*]) in a linear manner (Eqn 33). Highest values of Δ [CO₂*] and hence highest F_T at the beginning of the simulation period resulted in lowest E_A/R_S ratios (0.7) at this time (Figs 3b, 4a). However, this is an oversimplification of a complex interaction among several factors that influence Δ [CO₂*], such as transpiration rates, stem allometry and tapering, and respiratory activity (Hölttä & Kolari, 2009). Further research in xylem [CO₂] at different heights, in several species and across the season (Teskey & McGuire, 2007; Salomón *et al.*, 2016) would allow to better describe Δ [CO₂*] dynamics in mechanistic models.

Insights into the stem carbon balance

The stem C balance proposed in TReSpire (submodel C) reliably estimates the sink strength of physiological processes involved in the synthesis of new biomass, respiratory related costs and respiratory maintenance expenditures. However, quantification of C sources remains ambiguous due to high collinearity between parameters describing phloem unloading and SSI. Therefore, although helpful insights related to subdaily dynamics of stem C fluxes can be gained from this submodel, a more detailed description of C sources and more data for calibration would be necessary aiming at solving identifiability issues: A greater degree of complexity to describe and calibrate phloem transport and unloading could be incorporated by accounting for phloem leakage, sieve tubes properties, phloem sap viscosity, or different solutes not limited to sucrose equivalents (Thompson & Holbrook, 2003; Hölttä *et al.*, 2009; Minchin & Lacomte, 2017). Moreover, diameter variations in the upper stem could be recalibrated allowing vertical variation in R_{XY} , R_{OT} , C_{OT} and Φ (submodel A) to more

accurately estimate the vertical drop in turgor pressure. A more detailed description of SSI processes accounting for temporal dynamics in nonstructural carbohydrates (Hartmann & Trumbore, 2016) and osmoregulation (Lintunen *et al.*, 2016) would be necessary to more realistically represent starch metabolism, simplistically determined here by a target concentration of soluble sugars (Eqn 40). In this line, data on [SU] and [ST] at a finer temporal resolution (measured only once along the simulated period) would help to avoid numerical problems that arose during the estimation of the parameter covariance matrix. Finally, inclusion of woody tissue photosynthesis, intentionally impeded and neglected here to avoid interference with E_A measurements, would help to better quantify stem C sources given its significant impact on whole-plant C budgets (Ávila *et al.*, 2014; Steppe *et al.*, 2015).

Outlook

Ideally, biophysical models should be flexible enough to incorporate knowledge at the biochemical level, while organ-level outputs could be implemented within large-scale models. Firstly, growth and respiratory costs integrated in TReSpire (Eqns 18–22) could be itemised to subprocesses and biochemical reactions described in detail elsewhere (Amthor, 2000; Cannell & Thornley, 2000) and include new biochemical knowledge. Secondly, TReSpire (submodels A & B) could improve accuracy of current estimates of whole-plant respiration at larger spatial scales in the long term, as it is the first process-based model to predict R_S independently of leaf metabolism, practice commonly applied by TBMs. As a first step, TReSpire performance should be tested across different species and biomes, during longer periods encompassing at least one growing season, and along wide gradients of environmental conditions. If model calibration and simulation exhibit robustness, TReSpire output might be considered to be transferred to large-scale models. In this line, establishment of international research networks to collect and share data on stem respiratory fluxes (CO₂ efflux and/or O₂ influx) (see Yang *et al.*, 2016) and hydraulic measurements to run and calibrate TReSpire (submodels A and B) could result in a global database of stem respiratory traits (e.g. Y_G , Q_{10} and $R_{b,N}$). Such database would be useful to compare stem respiratory physiology and ecology of different species across spatial and temporal scales.

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Author contributions

RLS and KS planned and designed the research. RLS, LdR and JO performed the measurements. RLS developed TReSpire, analysed data and wrote the first draft of the manuscript. DJWDP and KS contributed to the writing of the manuscript.

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Supporting Information

Additional Supporting Information may be found online in the Supporting Information section at the end of the article.

Fig. S1 Thickness of the outer tissues (d_{OT}) measured with point dendrometers, and comparison of d_{OT} simulations using linear and nonlinear fits.

Fig. S2 Nonlinear allometric relationship between stem diameter (D) and thickness of the outer tissues (d_{OT}).

Fig. S3 Temporal variation in xylem CO₂ concentration in the gas phase (xylem [CO₂]) above and below a stem segment monitored in an ancillary maple tree, and the resulting vertical gradient ($\Delta X_{\text{ylem}} [\text{CO}_2]$); corresponding CO₂ concentration dissolved in the sap solution in the liquid phase (sap [CO₂*]) and the resulting vertical gradient ($\Delta \text{sap} [\text{CO}_2^*]$).

Methods S1 Measurements and sensors.

Methods S2 Experiment setup.

Notes S1 Details of TReSpire.

Table S1 Values of parameters implemented in TReSpire.

Table S2 Sensitivity analysis of submodel C.

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See also the Commentary on this article by Meir *et al.*, 225: 1824–1827.